Terraform and CDK (CDKTF)

Deployed AWS Infrastructure with Multiple Programming Python + Terraform CDKTF

PROBLEM

I wanted to create an AWS S3 bucket using **Python + CDKTF** instead of writing Terraform HCL. I also needed versioning, encryption, public access fizefsh, and proper tags.

HOW I DEBUGGED IT

First, I checked my AWS access. I checked the Python and package versions first.

Instead of changing the system Python,

I created a **virtual environment (venv)** and installed the required packages there.

Normal/system Python → you tried it → it did NOT work

Python venv → you tried it → it WORKED. Then I tested the Python code and ran

cdktf synth

After that, I ran:

cdktf diff to make sure everything looked right before deployment.

THE SOLUTION

```
cdktf deploy
```

The S3 bucket was created successfully with versioning, encryption, public access blocked, and proper tags.

Has anyone else faced this problem?

Have you used Python + CDKTF for AWS infrastructure?

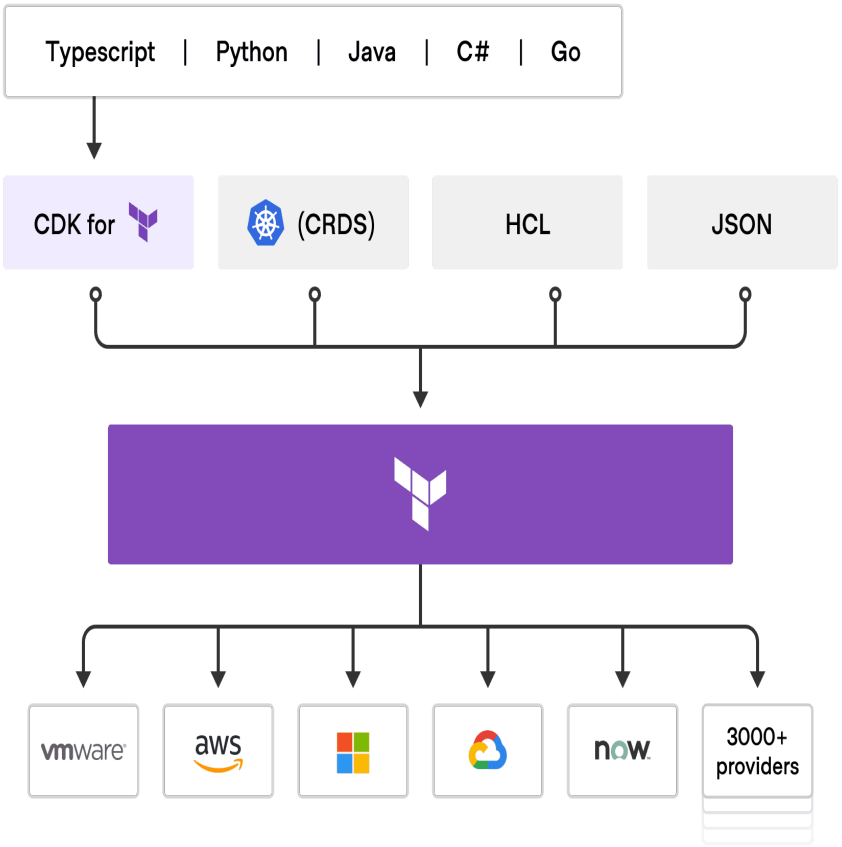
How did you fix it?

Note: All infrastructure images, official docs, and video links are included on the last page

Blog:https://vishnusai.hashnode.dev/terraform-and-cdk-cdktf?utm_source=hashnode&utm_medium=feed

HashnodProfile:https://hashnode.com/@vishnusaiK

GitHub : https://github.com/ksaivishnusaikvs



2. Prerequisites

Ubuntu EC2 instance, Python 3, pip, Node.js/npm, CDKTF CLI, Terraform CLI
AWS CLI
AWS credentials or an IAM role with permission to create the required S3 resources

Verify AWS Access

aws sts get-caller-identity

```
No outputs found.
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$
aws sts get-caller-identity
{
    "UserId": "AIDAR3O2KSZSEGMV2DJNG",
    "Account": "127696279140",
    "Arn": "arn:aws:iam::127696279140:user/vishnusai"
}
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$
```

Step 1: Create infrastructure: Create the CDKTF Project

cd ~
rm -rf prod-s3
mkdir prod-s3
cd prod-s3

python3 -m venv .venv
source .venv/bin/activate

```
  Synthesizing
ubuntu@ip-172-31-36-199:~/prod-s3$ cd ~
rm -rf prod-s3
mkdir prod-s3
cd prod-s3
ubuntu@ip-172-31-36-199:~/prod-s3$ python3 --version
pip3 --version
Python 3.12.3
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
```

pip install --upgrade pip
pip install cdktf constructs cdktf-cdktf-provider-aws

```
source .venv/bin/activate
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ pip install --upgrade pip
Requirement already satisfied: pip in ./.venv/lib/python3.12/site-packages (24.0)
Collecting pip
  Using cached pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.2.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
      Successfully uninstalled pip-24.0
Successfully installed pip-26.2.1
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ pip install cdktf constructs cdktf-cdktf-provider-aws
Collecting cdktf
  Downloading cdktf-0.21.0-py3-none-any.whl.metadata (1.3 kB)
Collecting constructs
  Downloading constructs-10.8.1-py3-none-any.whl.metadata (2.9 kB)
Collecting cdktf-cdktf-provider-aws
  Downloading cdktf_cdktf_provider_aws-21.22.1-py3-none-any.whl.metadata (4.1 kB)
Collecting jsii<2.0.0,>=1.112.0 (from cdktf)
  Downloading jsii-1.140.0-py3-none-any.whl.metadata (79 kB)
Collecting publication>=0.0.3 (from cdktf)
  Downloading publication-0.0.3-py2.py3-none-any.whl.metadata (8.7 kB)
Collecting typeguard<4.3.0,>=2.13.3 (from cdktf)
  Downloading typeguard-4.2.1-py3-none-any.whl.metadata (3.7 kB)
Collecting attrs<27.0,>=21.2 (from jsii<2.0.0,>=1.112.0->cdktf)
  Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting cattrs<27.0,>=1.8 (from jsii<2.0.0,>=1.112.0->cdktf)
  Downloading cattrs-26.2.0-py3-none-any.whl.metadata (8.5 kB)
Collecting typeguard<4.3.0,>=2.13.3 (from cdktf)
  Downloading typeguard-2.13.3-py3-none-any.whl.metadata (3.6 kB)
Collecting python-dateutil (from jsii<2.0.0,>=1.112.0->cdktf)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting typing_extensions<5.0,>=3.8 (from jsii<2.0.0,>=1.112.0->cdktf)
  Using cached typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting six>=1.5 (from python-dateutil->jsii<2.0.0,>=1.112.0->cdktf)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading cdktf-0.21.0-py3-none-any.whl (2.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.4/2.4 MB 109.0 MB/s  0:00:00
Downloading constructs-10.8.1-py3-none-any.whl (787 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 787.0/787.0 kB 46.4 MB/s  0:00:00
Downloading jsii-1.140.0-py3-none-any.whl (503 kB)
Downloading typeguard-2.13.3-py3-none-any.whl (17 kB)
Using cached attrs-26.1.0-py3-none-any.whl (67 kB)
Downloading cattrs-26.2.0-py3-none-any.whl (74 kB)
Using cached typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading cdktf_cdktf_provider_aws-21.22.1-py3-none-any.whl (59.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 59.3/59.3 MB 172.0 MB/s  0:00:00
Downloading publication-0.0.3-py2.py3-none-any.whl (7.7 kB)
```

STEP2: Create the CDKTF configuration

```
cat > cdktf.json <<'EOF'
{
  "language": "python",
  "app": "python3 main.py"
}
EOF
```

This tells CDKTF that the application entry point is main.py and that Python is the project language.

```
Requirement already satisfied: six>=1.5 in ./.venv/lib/python3.12/site-packages (from python-dateutil->jsii<2.0.0,>=1.112.0->cdktf) (1.17.0)
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cat > cdktf.json <<'EOF'
{
  "language": "python",
  "app": "python3 main.py"
}
EOF
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ nano main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 main.py
Traceback (most recent call last):
  File "/home/ubuntu/prod-s3/main.py", line 3, in <module>
    from cdktf_cdktf_provider_aws import AwsProvider
ImportError: cannot import name 'AwsProvider' from 'cdktf_cdktf_provider_aws' (/home/ubuntu/prod-s3/.venv/lib/python3.12/site-packages/cdktf_cdktf_provider_aws/__init__.py)
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cd ~/prod-s3
source .venv/bin/activate
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 -c "from cdktf_cdktf_provider_aws.provider import AwsProvider; print(AwsProvider)"
<class 'cdktf_cdktf_provider_aws.provider.AwsProvider'>
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 -c "from cdktf_cdktf_provider_aws.s3_bucket import S3Bucket; print(S3Bucket)"
<class 'cdktf_cdktf_provider_aws.s3_bucket.S3Bucket'>
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ nano main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ vi main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cdktf diff

  Synthesizing
```


```
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cat > cdktf.json <<'EOF'
{
  "language": "python",
  "app": "python3 main.py"
}
EOF
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ nano main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 main.py
Traceback (most recent call last):
  File "/home/ubuntu/prod-s3/main.py", line 3, in <module>
    from cdktf_cdktf_provider_aws import AwsProvider
ImportError: cannot import name 'AwsProvider' from 'cdktf_cdktf_provider_aws' (/home/ubuntu/prod-s3/.venv/lib/python3.12/site-packages/cdktf_cdktf_provider_aws/__init__.py)
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cd ~/prod-s3
source .venv/bin/activate
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 -c "from cdktf_cdktf_provider_aws.provider import AwsProvider; print(AwsProvider)"
<class 'cdktf_cdktf_provider_aws.provider.AwsProvider'>
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 -c "from cdktf_cdktf_provider_aws.s3_bucket import S3Bucket; print(S3Bucket)"
<class 'cdktf_cdktf_provider_aws.s3_bucket.S3Bucket'>
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ nano main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ vi main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cdktf diff
```

Step 3: Write code for infra Python Infrastructure Code. Create the infrastructure file

```
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cat main.py
from cdktf import App, TerraformStack
from constructs import Construct

from cdktf_cdktf_provider_aws.provider import AwsProvider
from cdktf_cdktf_provider_aws.s3_bucket import S3Bucket
from cdktf_cdktf_provider_aws.s3_bucket_versioning import S3BucketVersioningA
from cdktf_cdktf_provider_aws.s3_bucket_public_access_block import (
    S3BucketPublicAccessBlock
)
from cdktf_cdktf_provider_aws.s3_bucket_server_side_encryption_configuration import (
    S3BucketServerSideEncryptionConfigurationA
)


class ProdS3Stack(TerraformStack):

    def __init__(self, scope: Construct, stack_id: str):
        super().__init__(scope, stack_id)

        # AWS provider
        AwsProvider(
            self,
            "aws",
            region="ap-south-1"
        )

        # S3 bucket
        bucket = S3Bucket(
            self,
            "prod-bucket",
            bucket="kv-prod-s3-demo-20260909",
            tags={
                "Environment": "prod",
                "Application": "my-app",
                "ManagedBy": "cdktf"
            }
        )

        # Versioning
        S3BucketVersioningA(
            self,
            "versioning",
            bucket=bucket.id,
            versioning_configuration={
                "status": "Enabled"
            }
        )
```

```
                "ManagedBy": "cdktf"
            }
        )

        # Versioning
        S3BucketVersioningA(
            self,
            "versioning",
            bucket=bucket.id,
            versioning_configuration={
                "status": "Enabled"
            }
        )

        # Block public access
        S3BucketPublicAccessBlock(
            self,
            "public-access",
            bucket=bucket.id,
            block_public_acls=True,
            block_public_policy=True,
            ignore_public_acls=True,
            restrict_public_buckets=True
        )

        # Server-side encryption
        S3BucketServerSideEncryptionConfigurationA(
            self,
            "encryption",
            bucket=bucket.id,
            rule=[
                {
                    "apply_server_side_encryption_by_default": [
                        {
                            "sse_algorithm": "AES256"
                        }
                    ]
                }
            ]
        )

app = App()

ProdS3Stack(app, "prod-s3")

app.synth()
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$
```

CODE :

```
from cdktf import App, TerraformStack
from constructs import Construct

from cdktf_cdktf_provider_aws.provider import AwsProvider
from cdktf_cdktf_provider_aws.s3_bucket import S3Bucket
from cdktf_cdktf_provider_aws.s3_bucket_versioning import S3BucketVersioningA
from cdktf_cdktf_provider_aws.s3_bucket_public_access_block import (
        S3BucketPublicAccessBlock
)
from cdktf_cdktf_provider_aws.s3_bucket_server_side_encryption_configuration import (
    S3BucketServerSideEncryptionConfigurationA
)


class ProdS3Stack(TerraformStack):

        def __init__(self, scope: Construct, stack_id: str):
        super().__init__(scope, stack_id)

        # AWS provider
        AwsProvider(
        self,
        "aws",
        region="ap-south-1"
        )

        # S3 bucket
        bucket = S3Bucket(
        self,
        "prod-bucket",
       bucket="kv-prod-s3-demo-20260909",
        tags={
        "Environment": "prod",
        "Application": "my-app",
        "ManagedBy": "cdktf"
        }
        )

        # Versioning
        S3BucketVersioningA(
        self,
        "versioning",
        bucket=bucket.id,
        versioning_configuration={
        "status": "Enabled"
        }
        )

        # Block public access
        S3BucketPublicAccessBlock(
        self,
        "public-access",
        bucket=bucket.id,
        block_public_acls=True,
        block_public_policy=True,
        ignore_public_acls=True,
        restrict_public_buckets=True
        )
```

```
        # Server-side encryption
    S3BucketServerSideEncryptionConfigurationA(
        self,
        "encryption",
        bucket=bucket.id,
        rule=[
        {
              "apply_server_side_encryption_by_default": [
                  {
                    "sse_algorithm": "AES256"
                  }
                  ]
        }
        ]
        )


app = App()

ProdS3Stack(app, "prod-s3")

app.synth()
```

Step 4: What the Python Code Creates

AWS Provider: Uses ap-south-1 as the AWS region.

S3 Bucket: Creates the production bucket.

Tags: Adds Environment, Application, and ManagedBy tags for ownership and FinOps tracking.

Versioning: Keeps object versions so accidental changes/deletions can be recovered.

Public Access Block: Blocks public ACLs and public bucket policies.

Encryption: Enables S3 server-side encryption using AES256.

Step5: python3 main.py

Test the Python Code. If there is no Python traceback/error, the CDKTF application code is valid enough to synthesize.

```
Synthesizing
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cd ~/prod-s3
nano main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ vi main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ python3 main.py
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cdktf synth

Generated Terraform code for the stacks: prod-s3
```

Step6: cdktf synth

CDKTF converts the Python infrastructure definition into Terraform configuration under the generated CDKTF output.

```
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cdktf synth

Generated Terraform code for the stacks: prod-s3
```

Step7: cdktf diff

Review the resources before deployment. In the successful setup, the plan showed 4 resources to add and 0 to change or destroy.

```
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cdktf diff
prod-s3  Initializing the backend...
prod-s3
         Successfully configured the backend "local"! Terraform will automatically
         use this backend unless the backend configuration changes.
prod-s3  Initializing provider plugins...
         - Finding hashicorp/aws versions matching "6.25.0"...
prod-s3  - Installing hashicorp/aws v6.25.0...
prod-s3  - Installed hashicorp/aws v6.25.0 (signed by HashiCorp)

         Terraform has created a lock file .terraform.lock.hcl to record the provider
         selections it made above. Include this file in your version control repository
         so that Terraform can guarantee to make the same selections by default when
         you run "terraform init" in the future.
prod-s3  Terraform has been successfully initialized!

         You may now begin working with Terraform. Try running "terraform plan" to see
         any changes that are required for your infrastructure. All Terraform commands
         should now work.

         If you ever set or change modules or backend configuration for Terraform,
         rerun this command to reinitialize your working directory. If you forget, other
         commands will detect it and remind you to do so if necessary.
prod-s3  - Fetching hashicorp/aws 6.25.0 for linux_amd64...
prod-s3  - Retrieved hashicorp/aws 6.25.0 for linux_amd64 (signed by HashiCorp)
         - Obtained hashicorp/aws checksums for linux_amd64; All checksums for this platform were already tracked in the lock file
prod-s3  Success! Terraform has validated the lock file and found no need for changes.
prod-s3  Terraform used the selected providers to generate the following execution
         plan. Resource actions are indicated with the following symbols:
           + create

         Terraform will perform the following actions:
prod-s3    # aws_s3_bucket.prod-bucket (prod-bucket) will be created
           + resource "aws_s3_bucket" "prod-bucket" {
               + acceleration_status         = (known after apply)
               + acl                         = (known after apply)
               + arn                         = (known after apply)
               + bucket                      = "kv-prod-s3-demo-20260909"
               + bucket_domain_name          = (known after apply)
               + bucket_prefix               = (known after apply)
               + bucket_region               = (known after apply)
               + bucket_regional_domain_name = (known after apply)
               + force_destroy               = false
               + hosted_zone_id              = (known after apply)
               + id                          = (known after apply)
               + object_lock_enabled         = (known after apply)
               + policy                      = (known after apply)
               + region                      = "ap-south-1"
               + request_payer               = (known after apply)
```

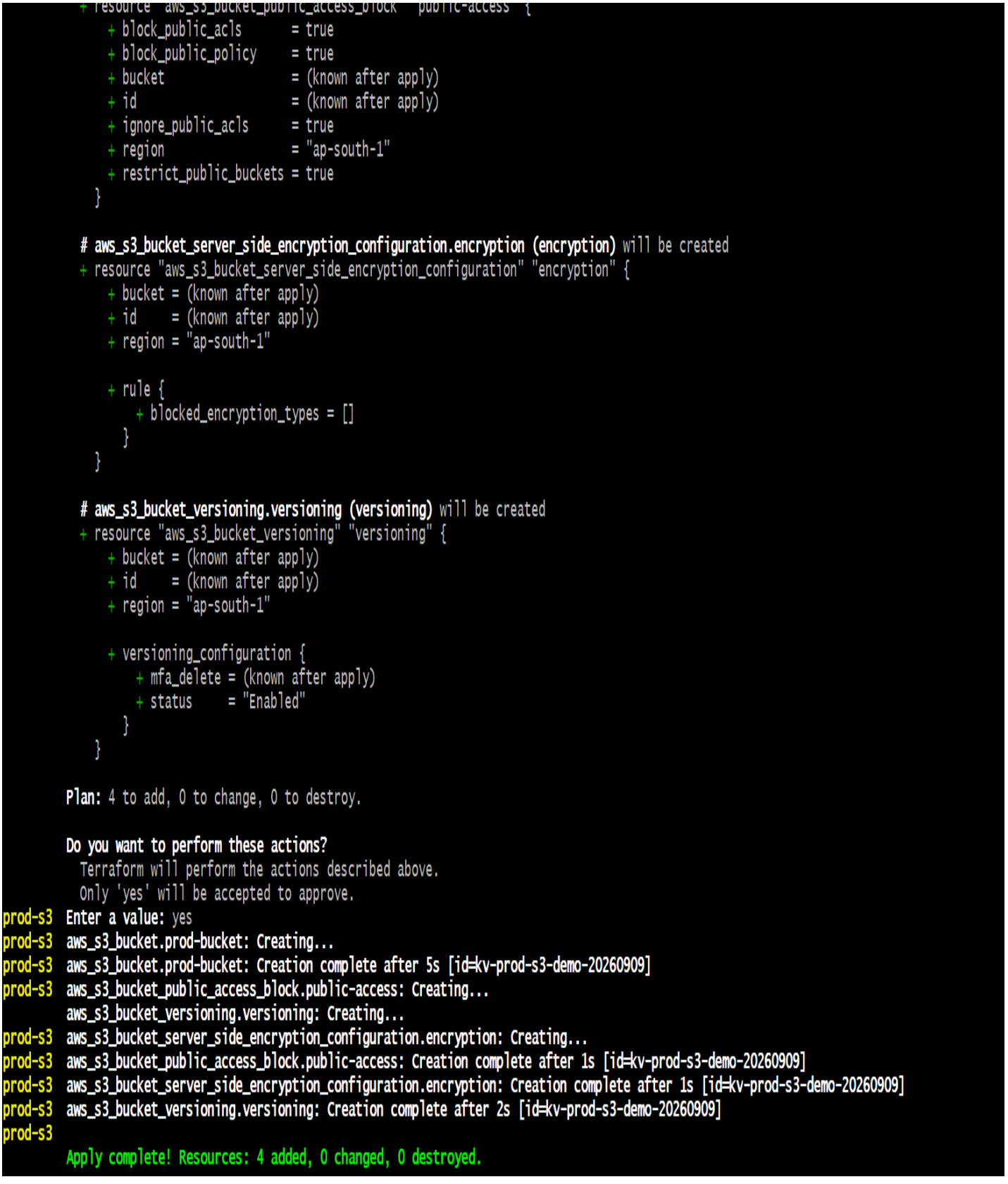
```
      + server_side_encryption_configuration (known after apply)

      + versioning (known after apply)

      + website (known after apply)
    }

  # aws_s3_bucket_public_access_block.public-access (public-access) will be created
  + resource "aws_s3_bucket_public_access_block" "public-access" {
      + block_public_acls       = true
      + block_public_policy     = true
      + bucket                  = (known after apply)
      + id                      = (known after apply)
      + ignore_public_acls      = true
      + region                  = "ap-south-1"
      + restrict_public_buckets = true
    }

  # aws_s3_bucket_server_side_encryption_configuration.encryption (encryption) will be created
  + resource "aws_s3_bucket_server_side_encryption_configuration" "encryption" {
      + bucket = (known after apply)
      + id     = (known after apply)
      + region = "ap-south-1"

      + rule {
          + blocked_encryption_types = []
        }
    }

  # aws_s3_bucket_versioning.versioning (versioning) will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = (known after apply)
      + id     = (known after apply)
      + region = "ap-south-1"

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 4 to add, 0 to change, 0 to destroy.

Saved the plan to: plan

To perform exactly these actions, run the following command to apply:
    terraform apply "plan"
```

Step8: cdktf deploy

When CDKTF asks for approval, choose Approve/yes. Terraform then creates the AWS resources

```
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ cdktf deploy
prod-s3  Initializing the backend...
prod-s3
prod-s3  Initializing provider plugins...
prod-s3  - Reusing previous version of hashicorp/aws from the dependency lock file
prod-s3  - Using previously-installed hashicorp/aws v6.25.0

         Terraform has been successfully initialized!

         You may now begin working with Terraform. Try running "terraform plan" to see
         any changes that are required for your infrastructure. All Terraform commands
         should now work.

         If you ever set or change modules or backend configuration for Terraform,
         rerun this command to reinitialize your working directory. If you forget, other
         commands will detect it and remind you to do so if necessary.
prod-s3  Terraform used the selected providers to generate the following execution plan.
         Resource actions are indicated with the following symbols:
           + create

         Terraform will perform the following actions:
prod-s3    # aws_s3_bucket.prod-bucket (prod-bucket) will be created
           + resource "aws_s3_bucket" "prod-bucket" {
               + acceleration_status         = (known after apply)
               + acl                         = (known after apply)
               + arn                         = (known after apply)
               + bucket                      = "kv-prod-s3-demo-20260909"
               + bucket_domain_name          = (known after apply)
               + bucket_prefix               = (known after apply)
               + bucket_region               = (known after apply)
               + bucket_regional_domain_name = (known after apply)
               + force_destroy               = false
               + hosted_zone_id              = (known after apply)
               + id                          = (known after apply)
               + object_lock_enabled         = (known after apply)
               + policy                      = (known after apply)
               + region                      = "ap-south-1"
               + request_payer               = (known after apply)
               + tags                        = {
                   + "Application" = "my-app"
                   + "Environment" = "prod"
                   + "ManagedBy"   = "cdktf"
                 }
               + tags_all                    = {
                   + "Application" = "my-app"
                   + "Environment" = "prod"
                   + "ManagedBy"   = "cdktf"
                 }
```

Step 9: See the image below; infra was created with Terraform/CDKTF using Python, and the data was validated

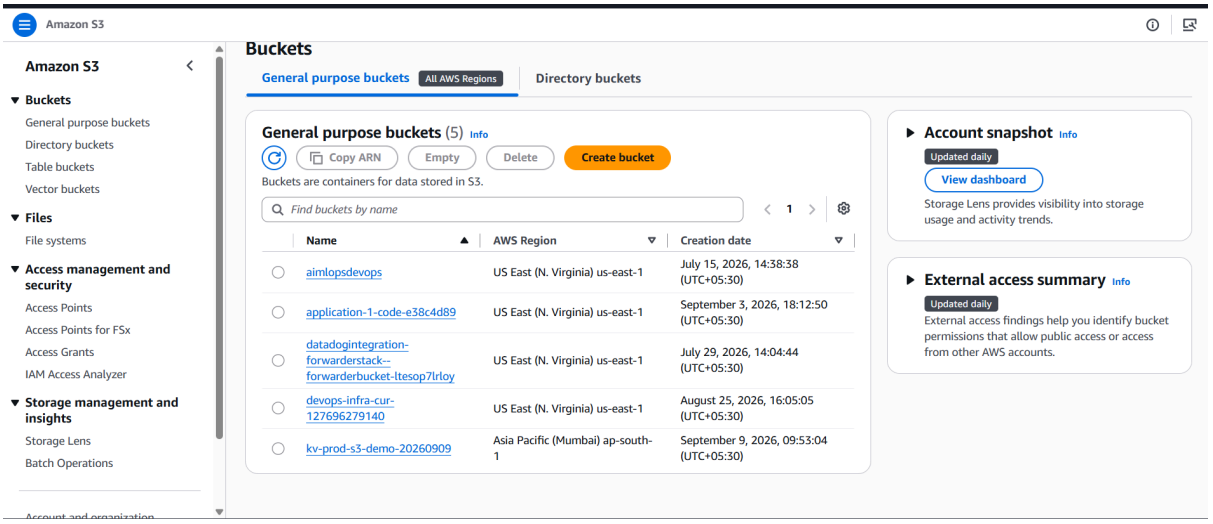


```
  + resource "aws_s3_bucket_public_access_block" "public-access" {
      + block_public_acls       = true
      + block_public_policy     = true
      + bucket                  = (known after apply)
      + id                      = (known after apply)
      + ignore_public_acls      = true
      + region                  = "ap-south-1"
      + restrict_public_buckets = true
    }

  # aws_s3_bucket_server_side_encryption_configuration.encryption (encryption) will be created
  + resource "aws_s3_bucket_server_side_encryption_configuration" "encryption" {
      + bucket = (known after apply)
      + id     = (known after apply)
      + region = "ap-south-1"

      + rule {
          + blocked_encryption_types = []
        }
    }

  # aws_s3_bucket_versioning.versioning (versioning) will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = (known after apply)
      + id     = (known after apply)
      + region = "ap-south-1"

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 4 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.
prod-s3  Enter a value: yes
prod-s3  aws_s3_bucket.prod-bucket: Creating...
prod-s3  aws_s3_bucket.prod-bucket: Creation complete after 5s [id=kv-prod-s3-demo-20260909]
prod-s3  aws_s3_bucket_public_access_block.public-access: Creating...
         aws_s3_bucket_versioning.versioning: Creating...
prod-s3  aws_s3_bucket_server_side_encryption_configuration.encryption: Creating...
prod-s3  aws_s3_bucket_public_access_block.public-access: Creation complete after 1s [id=kv-prod-s3-demo-20260909]
prod-s3  aws_s3_bucket_server_side_encryption_configuration.encryption: Creation complete after 1s [id=kv-prod-s3-demo-20260909]
prod-s3  aws_s3_bucket_versioning.versioning: Creation complete after 2s [id=kv-prod-s3-demo-20260909]
prod-s3
         Apply complete! Resources: 4 added, 0 changed, 0 destroyed.
```

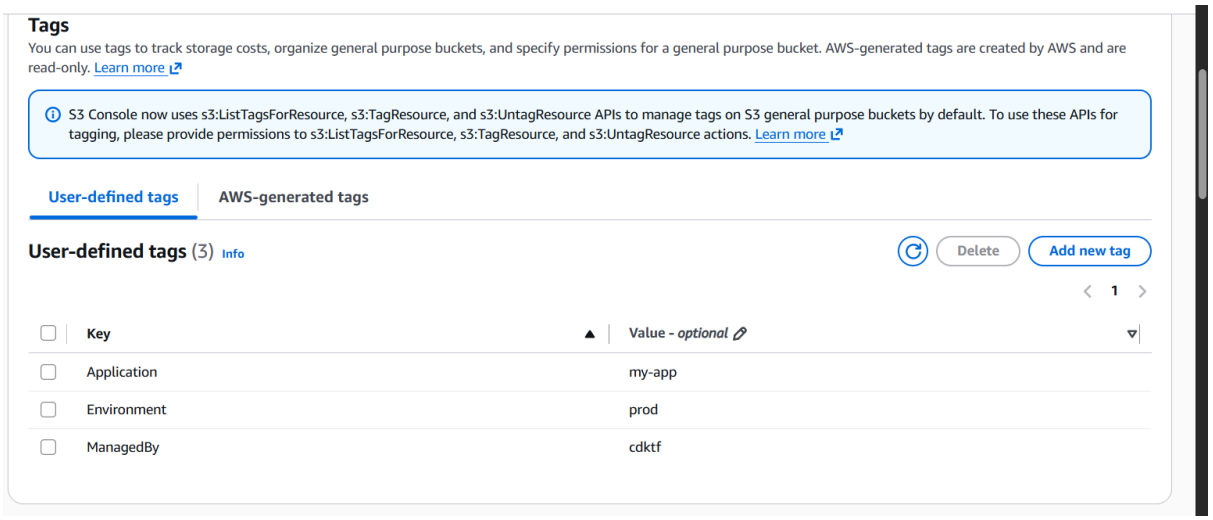
Step 10: Verify whether the S3 Bucket infra is created or not in the cloud are not state file create not in local

```
app.synth()
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ ls -la
total 32
drwxrwxr-x  4 ubuntu ubuntu 4096 Sep  9 04:23 .
drwxr-x--- 54 ubuntu ubuntu 4096 Sep  9 04:41 ..
drwxrwxr-x  5 ubuntu ubuntu 4096 Sep  9 03:55 .venv
-rw-rw-r--  1 ubuntu ubuntu  109 Sep  9 04:16 cdktf.json
drwxrwxr-x  3 ubuntu ubuntu 4096 Sep  9 04:18 cdktf.out
-rw-rw-r--  1 ubuntu ubuntu 2078 Sep  9 04:18 main.py
-rw-rw-r--  1 ubuntu ubuntu 6318 Sep  9 04:23 terraform.prod-s3.tfstate
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$ aws s3 ls
2026-07-15 09:08:38 aimlopsdevops
2026-09-03 12:42:50 application-1-code-e38c4d89
2026-07-29 08:34:44 datadogintegration-forwarderstack--forwarderbucket-ltesop7lrloy
2026-08-25 10:35:05 devops-infra-cur-127696279140
2026-09-09 04:23:04 kv-prod-s3-demo-20260909
(.venv) ubuntu@ip-172-31-36-199:~/prod-s3$
```

Step 11: Verify in the cloud ui the infrastructure resource is crete are not in cloud UI my s3 create throw CDKTF; name: s3 **kv-prod-s3-demo-20260909**



TAGS show the Terraform file; compare the two images



```
  # aws_s3_bucket_public_access_block.public-access (public-access) will be created
  + resource "aws_s3_bucket_public_access_block" "public-access" {
      + block_public_acls       = true
      + block_public_policy     = true
      + bucket                  = (known after apply)
      + id                      = (known after apply)
      + ignore_public_acls      = true
      + region                  = "ap-south-1"
      + restrict_public_buckets = true
    }

  # aws_s3_bucket_server_side_encryption_configuration.encryption (encryption) will be created
  + resource "aws_s3_bucket_server_side_encryption_configuration" "encryption" {
      + bucket = (known after apply)
      + id     = (known after apply)
      + region = "ap-south-1"

      + rule {
          + blocked_encryption_types = []
        }
    }

  # aws_s3_bucket_versioning.versioning (versioning) will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = (known after apply)
      + id     = (known after apply)
      + region = "ap-south-1"

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 4 to add, 0 to change, 0 to destroy.
```

Objects | Metadata | **Properties** | Permissions | Metrics | Management | File systems | Access Points

Bucket overview

AWS Region
Asia Pacific (Mumbai) ap-south-1

Amazon Resource Name (ARN)
arn:aws:s3:::kv-prod-s3-demo-20260909

Creation date
September 9, 2026, 09:53:04 (UTC+05:30)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more

Bucket Versioning
Enabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more
Disabled

Objects | Metadata | Properties | **Permissions** | Metrics | Management | File systems | Access Points

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about How IAM analyzer findings work
View analyzer for ap-south-1

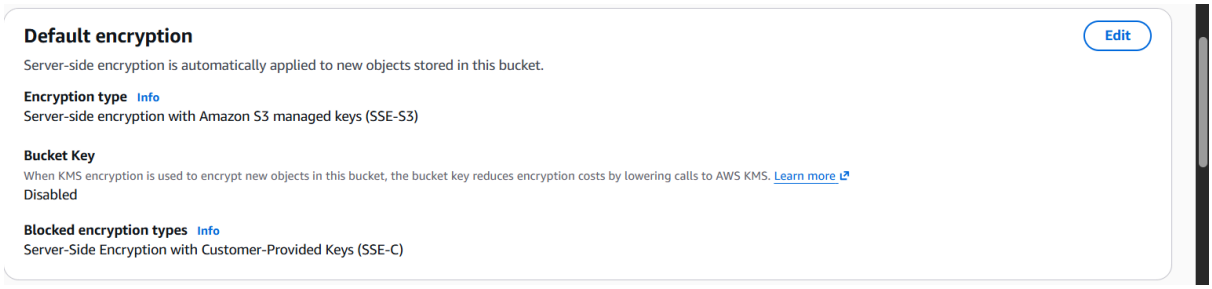
Block public access (bucket settings)

Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

Block *all* public access
On

▶ **Individual Block Public Access settings for this bucket**



REFDOC:

Hashicorp Ofiical : https://developer.hashicorp.com/terraform/cdktf

AWS: https://aws.amazon.com/blogs/opensource/announcing-cdk-for-terraform-on-aws/

installsteps: 1:https://spacelift.io/blog/terraform-cdk

2:https://dev.to/aws-builders/cdk-for-terraform-cdktf-on-aws-how-to-configure-an-s3-remote-backend-and-deploy-a-lambda-function-url-using-python-3okk

CDKTF official HashiCorp channel: https://www.youtube.com/watch?v=Y9suaDVT-SY&t=27s

https://developer.hashicorp.com/terraform/tutorials/cdktf/cdktf-install

DOCsteps: https://spacelift.io/blog/terraform-cdk